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Automatic Bloom's Taxonomy Question Generation Using RAG and ModernBERT Classification with Regenerative Feedback

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ABSTRACT Automatic question generation has emerged as a key enabler of scalable educational assessment, yet existing approaches face two persistent limitations. Large Language Models alone produce questions without guaranteed alignment to a specific cognitive level, and prior studies have either treated cognitive classification of existing question banks in isolation from generation, or relied on coarse difficulty labels without explicit mapping to Bloom's Taxonomy. This study proposes an integrated framework combining Retrieval Augmented Generation, four open weight Large Language Models, and the pretrained ModernBERT Bloom Taxonomy classifier within a unified pipeline with a regenerative feedback loop. RAG retrieves contextually relevant passages from a curated OpenStax knowledge base spanning five academic subjects, the LLMs generate candidate questions conditioned on a target Bloom's level from Remember (C1) through Create (C6), and the ModernBERT classifier verifies the cognitive level of each generated question. When the predicted level deviates from the target, the feedback loop iteratively refines the output within a maximum of three attempts. Llama-3.1-8B, Qwen-3-8B, Mistral-7B, and Gemma-3-4B were evaluated under identical 4-bit precision via Ollama runtime on 1,914 questions per model spanning 319 topics. The ModernBERT classifier achieved 86.36 percent accuracy on the Kaggle Bloom's Taxonomy dataset, while Gemma-3-4B attained the highest end to end accuracy of 79.26 percent and Llama-3.1-8B obtained the strongest overall F1 score of 0.8817, leading at four of six Bloom's levels. The results confirm that classifier based verification combined with regenerative feedback substantially improves cognitive level alignment.

INDEX TERMS Automatic Question Generation, Bloom's Taxonomy, Large Language Models, ModernBERT, Retrieval-Augmented Generation.

I. INTRODUCTION

The development of artificial intelligence (AI) technology has brought significant changes in the field of education, particularly in the development of digital-based learning evaluation systems [1] [2]. One of the main challenges in the learning process is the preparation of questions that are not only relevant to the material but also appropriate to the cognitive abilities of the learners. Bloom's Taxonomy is a framework commonly used to classify levels of thinking ability, ranging from lower-order thinking skills such as Remember (C1) to higher-order thinking skills such as Create (C6) [3]. Ensuring that generated questions are appropriately mapped to these six cognitive levels remains a non-trivial challenge, particularly when done automatically at scale.

The emergence of Large Language Models (LLMs) and transformer-based architectures has opened new possibilities for automating educational content generation. However, LLMs alone tend to produce questions without guaranteed alignment to a specific cognitive level, as they lack an explicit verification mechanism to ensure the generated output conforms to a defined Bloom's Taxonomy level [4]. This limitation motivates the integration of a dedicated classifier to provide post-generation cognitive-level verification.

Previous research employs Retrieval-Augmented Generation (RAG) with Large Language Models (LLMs) to automatically generate questions from knowledge bases [5]. The results indicate that models such as Llama-3-8B, GPT-2, and Cohere can produce questions with high accuracy, but still face limitations in maintaining consistent difficulty levels and stable question quality, and do not utilize Bloom's Taxonomy to classify question difficulty. Other studies have explored the application of machine learning classifiers to categorize questions according to Bloom's Taxonomy levels, yet these approaches operate independently of a generation pipeline and rely on fixed, manually curated question banks [6].

Despite their contributions, prior studies remain limited because the first focuses only on cognitive classification without generating new questions, while the second generates questions but does not align difficulty with Bloom's Taxonomy C1–C6. In addition, RAG-based studies generally apply only broad difficulty categories such as low, medium, and high without explicit cognitive mapping.

A critical component that addresses the cognitive classification gap is the availability of pre-trained transformer-based classifiers fine-tuned specifically for Bloom's Taxonomy. This study leverages ModernBERT-Bloom-Taxonomy [7], a publicly available model on Hugging Face (manzarimalik/ModernBERT-Bloom-Taxonomy) that was fine-tuned from answerdotai/ModernBERT-large on a structured Bloom's Taxonomy dataset [8]. ModernBERT-large itself represents a significant modernization of the classical BERT encoder architecture, introducing Rotary Positional Embeddings (RoPE) for extended sequence support up to 8,192 tokens, alternating local-global attention mechanisms for efficient long-context processing, GeGLU activations, and Flash Attention with unpadding for hardware-optimized inference [9]. Pre-trained on two trillion tokens of English and code data, ModernBERT provides a substantially stronger representational foundation compared to its predecessor, making it well-suited for fine-grained text classification tasks such as cognitive-level prediction across the six Bloom's Taxonomy levels (BT1–BT6). The use of this pre-trained classifier eliminates the need to train a Bloom's classifier from scratch and provides a reliable, reproducible cognitive verification component within the proposed pipeline.

This study proposes an AI-based framework integrating Retrieval-Augmented Generation (RAG), four open-weight Large Language Models (LLMs), and the ModernBERT-Bloom-Taxonomy pretrained classifier to automatically generate questions with verified cognitive levels based on Bloom's Taxonomy. This integrated approach enables the production of contextually relevant and cognitively verifiable assessment items, supporting adaptive learning in Learning Management Systems (LMS). Accordingly, the contributions of this study include:

- Integration of RAG, LLMs, and a pre-trained ModernBERT-based classifier for automated question generation with cognitive-level verification.
- Explicit classification of question difficulty using Bloom's Taxonomy C1–C6 instead of general difficulty labels
- A regenerative feedback mechanism that iteratively refines generated questions to achieve cognitive-level alignment.
- Comprehensive comparative evaluation of four open-weight LLMs on cognitive-level alignment, providing empirical guidance on model selection for Bloom's Taxonomy question generation.

The remainder of this paper is organized as follows. Section II reviews related work on automatic question generation, Bloom's Taxonomy classification, and RAG-based approaches. Section III presents the proposed methodology, including system architecture, dataset preparation, RAG retrieval process, LLM-based question generation, and the ModernBERT-based classification of cognitive levels. Section IV describes the experimental setup, evaluation metrics, and performance results, including analysis of accuracy, per-class precision/recall/F1, and cognitive-level consistency via confusion matrix analysis. Section V concludes the study and outlines potential directions for future research, particularly in improving model robustness and expanding domain adaptability.

II. RELATED WORK

This section reviews three interconnected research streams that form the foundation of the proposed framework: (A) automatic question generation using Large Language Models and RAG-based approaches, (B) Bloom's Taxonomy cognitive-level classification, and (C) transformer-based encoder models for educational text classification. A summary of the reviewed studies and their respective capabilities is presented in Table 1, which serves as the basis for identifying the research gap addressed by this work.

A. AUTOMATIC QUESTION GENERATION WITH RAG

Automatic Question Generation (AQG) has been a long-standing challenge in educational AI, evolving from early rule-based and template-driven approaches toward deep learning and generative models [10]. With the emergence of Large Language Models (LLMs), AQG systems can now produce contextually coherent questions from unstructured text with minimal human supervision.

Aisyah et al. [5] conducted a comparative study on RAG-based question generation using multiple LLMs including Llama-3-8B, GPT-2, and Cohere, evaluating their performance in generating questions from domain-specific knowledge bases. Their system retrieved relevant passages using vector similarity search and prompted the LLMs to generate questions accordingly. The study demonstrated that RAG substantially reduces hallucinations compared to standalone LLM generation, and that Llama-3-8B achieved competitive performance in terms of relevance and fluency. However, the authors noted persistent limitations in maintaining consistent difficulty levels across generated questions. Critically, no mechanism was employed to classify

TABLE 1. Comparison of related studies

Study	Method	Question Generation	Bloom's Taxonomy	Cognitive Verification	Feedback Loop	RAG
Aisyah et al. [5] (2024)	RAG + Llama-3-8B / GPT-2 / Cohere	✓	✗	✗	✗	✓
Banujan et al. [6] (2023)	ANN + LSTM + BERT	✗	✓ (prompt-guided)	✓ (classifier only)	✗	✗
Maity et al. [11] (2025)	RAG + ICL + GPT-4 / BART	✓	✗	✗	✗	✓
Scaria et al. [12] (2024)	Prompting + LLMs (GPT-4, Mistral, Llama)	✓	✓ (prompt-guided)	✗ (no classifier)	✗	✗
Kumar et al. [13] (2025)	ML / RNN / BERT / RoBERTa / LLM	✗	✓ (C1–C6)	✓ (classifier only)	✗	✗
Warner et al. [9] (2024)	ModernBERT pre-training	✗	✗	✗	✗	✗
Proposed Work	RAG + 4 LLM Models + ModernBERT-Bloom	✓	✓ (C1–C6)	✓ (classifier)	✓	✓

or verify the cognitive complexity of generated outputs against an established educational taxonomy

Maity et al. [11] explored In-Context Learning (ICL), RAG, and a hybrid RAG-ICL approach for AQG using GPT-4 and BART. Their hybrid model consistently outperformed individual methods in contextual accuracy and question relevance. Despite this, the study relied solely on automated metrics such as ROUGE and BLEU for evaluation, without assessing alignment to any cognitive framework. The absence of explicit cognitive-level targeting limits the pedagogical utility of the generated questions, particularly for structured assessments in Learning Management Systems (LMS).

Scaria et al. [12] examined the ability of five instruction fine-tuned LLMs including GPT-4, Mistral, and Llama variants to generate educational questions at different Bloom's Taxonomy levels using advanced prompting strategies. Their findings confirmed that LLMs can generate high-quality questions when prompted with adequate cognitive-level information, but exhibited significant variance in skill alignment, particularly for higher-order cognitive levels (C4–C6). The study also showed that automated evaluation metrics are insufficient for verifying cognitive alignment, highlighting the need for a dedicated classifier as a post-generation verification mechanism.

Collectively, these studies demonstrate the strengths of RAG in grounding question generation to relevant content while exposing a shared limitation: the absence of a closed-loop mechanism to verify and enforce cognitive-level alignment. The present work directly addresses this gap by integrating a pre-trained Bloom's Taxonomy classifier into the generation pipeline and incorporating a regenerative feedback loop that iteratively refines non-ideal outputs.

B. BLOOM'S TAXONOMY CLASSIFICATION

Parallel to generation research, a distinct body of work focuses on the automatic classification of existing questions according to Bloom's Taxonomy cognitive levels. These studies treat classification as a standalone task applied to pre-existing question banks, without coupling it to a generation component.

Banujan et al. [6] developed an automated classification system for undergraduate examination questions using deep learning models, specifically Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM) networks combined with word embedding techniques including GloVe, BERT, and TF-IDF. Their study collected 16,584 questions from multiple Sri Lankan universities and applied the Revised Bloom's Taxonomy for labeling. Results showed that the LSTM+BERT model significantly outperformed ANN+TF-IDF and LSTM+GloVe configurations based on ANOVA with Bonferroni correction, demonstrating the superiority of contextual embeddings for cognitive-level prediction. However, the system operates entirely on existing, manually curated question banks and provides no mechanism for generating new questions. The classification component is therefore decoupled from the educational content retrieval process.

Kumar et al. [13] conducted a systematic comparison of classification paradigms for Bloom's Taxonomy across Naive Bayes, Logistic Regression, SVM, LSTM, BiLSTM, GRU, BERT, RoBERTa, and LLM-based zero/few-shot approaches on a dataset of 600 labeled sentences. Their analysis identified research gaps in the existing literature: first, most studies rely on conventional ML with surface-level features; second, the effectiveness of deep learning on small datasets remains underexplored; and third, LLM-based zero-shot classification

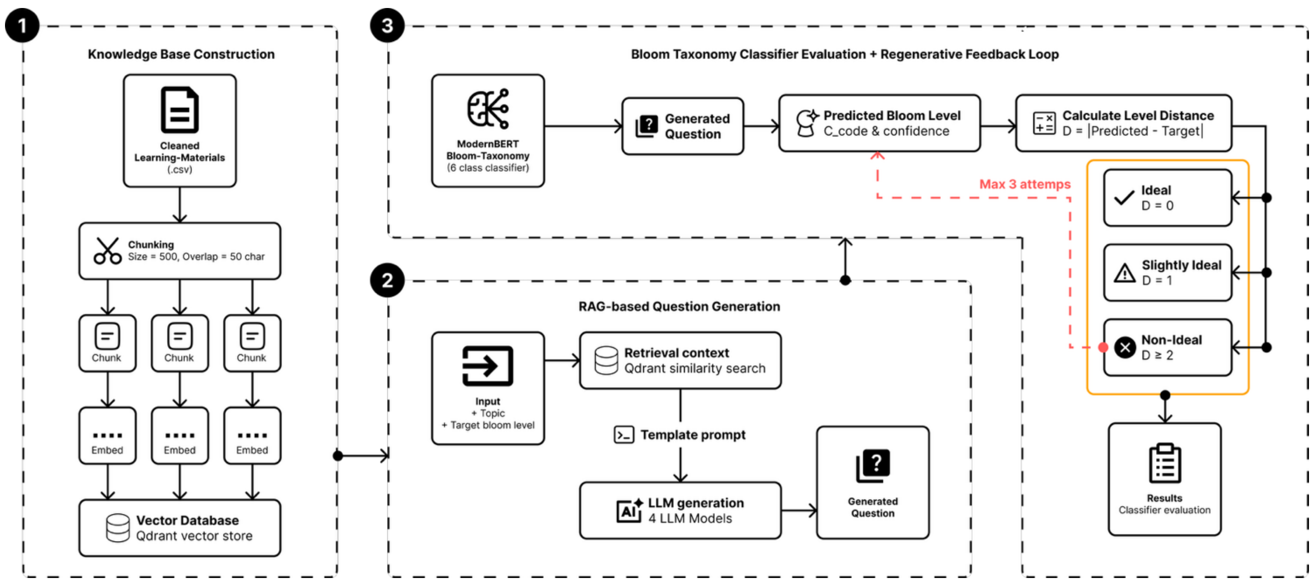

FIGURE 1. **Research Method**

introduces new opportunities not yet systematically benchmarked. While their work provides a comprehensive benchmark, it confirms that classification-only approaches do not address the need for integrated cognitive-level question generation.

These studies collectively establish that transformer-based classifiers, particularly BERT-family models, represent the current state of the art in Bloom's Taxonomy classification. However, they operate independently of question generation pipelines and do not provide a mechanism for producing new assessment content aligned to target cognitive levels.

C. TRANSFORMER-BASED ENCODERS

The backbone of the cognitive-level classifier in the proposed framework is ModernBERT, a recently introduced encoder-only transformer that represents a significant architectural advancement over classical BERT models [14]. Warner et al. [9] introduced ModernBERT as a Pareto improvement over prior encoders, trained on 2 trillion tokens of English and code data with a native context length of up to 8,192 tokens. The architecture incorporates Rotary Positional Embeddings (RoPE) for extended sequence support, alternating local-global attention patterns for efficient long-context processing, GeGLU activations, and Flash Attention with input unpadding for hardware-optimized inference. ModernBERT achieves state-of-the-art results on diverse classification and retrieval benchmarks, outperforming both base and large variants of RoBERTa, DeBERTa, and prior BERT checkpoints while maintaining significantly lower memory consumption and faster inference speeds.

Building on this foundation, Malik [7] fine-tuned ModernBERT-large on a structured Bloom's Taxonomy dataset (manzarimalik/ModernBERT-Bloom-Taxonomy, Hugging Face), yielding a six-class text classifier mapping educational questions to BT1 (Remember) through BT6 (Create). The model was fine-tuned using the Unsloth training framework and Hugging Face's TRL library, achieving 2x faster fine-tuning compared to standard procedures. Released under an Apache 2.0 license, this model provides a plug-and-play cognitive verification component that eliminates the need to train a Bloom's classifier from scratch. The availability of this pre-trained model is a key enabler of the proposed framework, as it allows cognitive-level verification to be performed efficiently within a real-time generation-evaluation loop.

III. PROPOSED METHOD

The proposed framework integrates three core components: Retrieval-Augmented Generation (RAG), multiple Large Language Models (LLMs), and a pre-trained Bloom's Taxonomy classifier into a unified pipeline for automated cognitive-level question generation. As illustrated in Fig. 1, the system operates across three sequential phases.

A. DATASET

This study utilizes two distinct datasets, each serving a different purpose within the proposed framework. The first is a knowledge base dataset constructed from OpenStax learning materials, used to support the RAG-based question generation pipeline [15]. The second is a Bloom's Taxonomy classification dataset sourced from Kaggle, used to evaluate the cognitive-level classification performance of the pre-trained ModernBERT-Bloom-Taxonomy classifier [8].

OpenStax was selected as the data source due to its structured, academically rigorous content spanning multiple disciplines, its free availability, and its permissive license for research use. Five subjects are included in the dataset: Biology, Physics, Chemistry, Psychology, and Economics. Learning materials are extracted from the corresponding OpenStax textbook PDF for each subject, producing a four-column structured dataset with the fields content, topic, subject, and keywords. Table 2 presents the top five entries of

TABLE 2. **Preprocessed OpenStax knowledge base dataset**

No	Content	Topic	Subject	Keywords
1	Genetics is the study of	Introduction to Patterns of Inheritance	Biology	introduction chapter…
2	Today, members of this plant …	Speciation	Biology	species, supercontinent…
3	In both prokaryotes ….	Transcription	Biology	dna, learning objectives …
4	Another scenario in which ….	Evidence of Evolution	Biology	population, founder effect …
5	Like terrestrial biomes …	Aquatic and Marine Biomes	Biology	wildlife refuge, national wildlife ….

the preprocessed OpenStax dataset as a representative sample of the knowledge base structure.

In addition to the OpenStax knowledge base, this study employs a publicly available Bloom's Taxonomy classification dataset. The dataset consists of 8,767 labeled educational questions spanning six cognitive levels defined by Bloom's Taxonomy: BT1 (Remember), BT2 (Understand), BT3 (Apply), BT4 (Analyze), BT5 (Evaluate), and BT6 (Create). Each entry contains two fields: a Questions column containing the question text, and a Category column containing the corresponding Bloom's Taxonomy level label. The questions originate from diverse academic domains and vary in length and linguistic complexity across cognitive levels. Table 3 presents the top five entries of the dataset as a representative sample of its structure.

TABLE 3. **Bloom's taxonomy classification dataset**

Questions	Category
About what proportion of the population of the US is living on farms?	BT 1
Correctly label the brain lobes indicated on the diagram below.	BT 1
Define compound interest.	BT 1
Define four types of traceability.	BT 1
Define mercantilism.	BT 1

B. PREPROCESSING

The OpenStax knowledge base dataset undergoes a two-stage preprocessing pipeline: content extraction and text cleaning, as illustrated in Fig. 2. The goal of this pipeline is to transform raw PDF-extracted content into clean, structured text suitable for embedding and retrieval in the RAG pipeline.

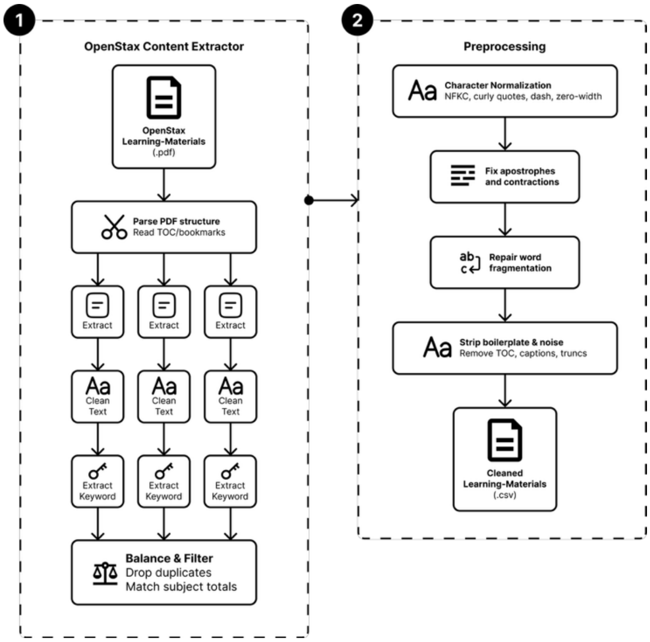

FIGURE 2. **OpenStax Two-stage Preprocessing Pipeline**

In the content extraction stage, each OpenStax textbook PDF is downloaded programmatically and parsed using the document's TOC and bookmark hierarchy. Sections at depth levels 1–4 are retained as candidate entries. Each section is subject to quality filtering: entries with fewer than 300 characters are discarded, entries exceeding 5,000 characters are truncated at the last complete sentence, and non-content sections such as Exercises, Glossary, and References are removed. Keywords are extracted using YAKE [16], retaining the top five bigram keywords per section. A stratified sampling procedure is then applied to equalize the number of entries across all five subjects, with duplicates removed prior to sampling.

Following extraction, the dataset undergoes a six-step text cleaning pipeline to address PDF extraction artefacts. Step 1 normalizes characters to NFKC Unicode form, standardizes punctuation marks, removes zero-width characters, and collapses whitespace. Step 2 restores apostrophes dropped during extraction using rule-based pattern matching for contractions and possessives. Step 3 repairs word fragmentation artefacts caused by PDF multi-column layout using a predefined correction dictionary. Step 4 removes boilerplate TOC prefixes, trailing chapter outline suffixes, and image credit annotations. Step 5 removes inline figure captions, contextual section markers, and figure references. Step 6 repairs entries truncated at the character boundary by trimming to the last complete sentence, provided the terminal punctuation occurs beyond 50% of the total text length.

C. KNOWLEDGE BASE CONSTRUCTION

The preprocessed OpenStax dataset is loaded as a collection of LangChain Document objects, with each row's content field serving as the document text and the topic, subject, and keywords fields preserved as metadata. Each document is subsequently split into fixed-size chunks using a RecursiveCharacterTextSplitter with a chunk size of 500 characters and an overlap of 50 characters, using the separator hierarchy ["\n\n", "\n", ". ", " ", ""] to ensure splits occur at natural linguistic boundaries where possible. Topic and subject metadata are propagated to each chunk to support filtered retrieval.

Each chunk is encoded into a 384-dimensional dense vector using the all-MiniLM-L6-v2 sentence embedding model [17], a lightweight transformer-based encoder pre-trained on large-scale sentence-pair datasets for semantic similarity tasks. The embedding of a chunk d is defined as Equation (1).

$$e(d) = Encoder(d), e(d) \in \mathbb{R}^{384} \quad (1)$$

All embeddings are L2-normalized, such that $\|e(d)\| = 1$. The encoded vectors are stored in an in-memory Qdrant vector database [18], configured with cosine distance as the similarity metric. Cosine similarity between a query vector $e(q)$ and a document vector $e(d)$ is computed as Equation (2).

$$sim(q, d) = \frac{(e(q) \cdot e(d))}{(\|e(q)\| \cdot \|e(d)\|)} \quad (2)$$

Since all vectors are L2-normalized, the cosine similarity computation reduces to the inner product $e(q) \cdot e(d)$. Given an input query q, the retriever returns the three document chunks with the highest semantic similarity scores from the vector database. These retrieved chunks are subsequently used as contextual information for the RAG-based question generation process. The resulting retriever R forms the document supply component of the RAG pipeline described in Section E.

D. BLOOM'S TAXONOMY CLASSIFIER

The cognitive-level classifier used in this study is ModernBERT-Bloom-Taxonomy, a publicly available model on Hugging Face fine-tuned from answerdotai/ModernBERT-large on the Bloom's Taxonomy classification dataset described in Section 3.2. ModernBERT-large is an encoder-only transformer pre-trained on two trillion tokens of English and code data with a native context length of 8,192 tokens. Its architecture incorporates Rotary Positional Embeddings (RoPE) for extended sequence support, alternating local-global attention for efficient long-context processing, GeGLU activations, and Flash Attention with input unpadding for hardware-optimized inference [9].

The fine-tuned model performs six-class text classification, mapping an input question to one of the Bloom's Taxonomy cognitive levels. The label mapping is defined in Table 4.

TABLE 4. Bloom's taxonomy label mapping

Label	Cognitive Level	C-Code
BT1	Remember	C1
BT2	Understand	C2
BT3	Apply	C3
BT4	Analyze	C4
BT5	Evaluate	C5
BT6	Create	C6

The classifier is loaded as a HuggingFace text-classification pipeline and operates on the contextual representation of the [CLS] token produced by the ModernBERT encoder. Given an input question text x, the model produces a hidden representation h_[CLS] ∈ $\mathbb{R}^d$, which is passed through a linear classification head followed by a softmax activation as Equation (3).

$$P_{(c|x)} = softmax(W \cdot h_{[CLS]} + b), \quad c \in \{BT1, BT2, BT3, BT4, BT5, BT6\} \quad (3)$$

Where W ∈ $\mathbb{R}^6$ is the weight matrix and b ∈ $\mathbb{R}^6$ is the bias vector of the classification head. The predicted Bloom's Taxonomy level is then defined as Equation (4).

$$\hat{y} = argmax_c P_{(c \mid x)} \quad (4)$$

The classifier also returns the confidence score of the predicted class, defined as Equation (5).

$$conf(\hat{y}) = max_c P_{(c \mid x)} \quad (5)$$

This model is loaded once at system initialization and reused across all classification calls in Phase 3, eliminating the overhead of repeated model loading during the feedback loop.

E. RAG-BASED QUESTION GENERATION

Question generation is performed by a RAG chain that combines the Qdrant retriever constructed in Phase 1A with a Large Language Model (LLM). RAG is an AI technique that enhances large language models by retrieving relevant information from external knowledge sources before generating a response [19]. The selected models, Llama-3.1-8B, Mistral-7B, Qwen-3-8B, and Gemma-3-4B, represent prominent open-weight LLM families with competitive performance in instruction-following, reasoning, and language understanding tasks, while remaining suitable for local deployment on consumer-grade hardware [20], [21]. This selection enables a comparative evaluation of cognitive-level question generation across models developed by different organizations. All models were deployed locally using Ollama [22] and executed on an NVIDIA GeForce RTX 3060 GPU with a temperature setting of 0.7.

The generation process takes two inputs: a topic τ and a target Bloom's Taxonomy level $L^* \in$ *{Remember, Understand, Apply, Analyze, Evaluate, Create}*. The retriever is invoked with the topic as the query to obtain the *top-k = 3* most relevant chunks from the Qdrant vector store. The retrieved chunks are concatenated into a single context string *C*, this computed as Equation (6).

$$C = d_1 \oplus d_2 \oplus d_3, \quad where\ d_i \in R(\tau) \qquad (6)$$

Where $\oplus$ denotes string concatenation separated by a section delimiter. The context *C*, topic τ, and target level L^* are injected into a structured prompt template. The prompt is designed to guide the LLM in generating questions whose cognitive complexity strictly conforms to the target Bloom's level. The template prompt follows the structure in Table 5.

TABLE 5. Template prompt for generating questions

LLM Prompt
"You are an expert educational-assessment designer. Generate [num_questions] exam question(s) at the [target_level] level of Bloom's Taxonomy (Revised). Use the following action verbs as cognitive anchors: [action_verbs]. The question must be directly answerable from the retrieved context and must not introduce information beyond it. Context: [C]. Topic: [τ]."

To enforce cognitive-level specificity, the prompt supplies a distinct set of action verbs for each Bloom's level as generation constraints, as summarized in Table 6. The action verbs used as generation constraints are adapted from the revised Bloom's Taxonomy framework and commonly used Bloom verb inventories in educational assessment design [23].

TABLE 6. Action verbs for each Bloom's level

Bloom's Level	Action Verbs
C1	define, list, identify, name, recall, state
C2	describe, explain, summarize, classify, compare
C3	apply, demonstrate, solve, use, compute, illustrate
C4	analyze, contrast, examine, differentiate, infer
C5	assess, critique, judge, defend, justify, evaluate
C6	design, construct, develop, formulate, propose, plan

Based on the prompt configuration and cognitive-level constraints described above, the question generation process is formalized in Equation (7).

$$Q = LLM_{(prompt(C,\tau,L*))}, \quad temp = 0.7 \qquad (7)$$

The LLM output is parsed by a string output parser that extracts individual question strings from the raw text, retaining only lines that contain a question mark as valid question candidates. The generated question Q is subsequently passed to Phase 3 for cognitive-level verification and feedback evaluation.

F. REGENERATIVE FEEDBACK LOOP

Upon generation of a candidate question Q from Phase 2, the ModernBERT-Bloom-Taxonomy classifier and is applied to predict the cognitive level of the generated question. The predicted level $\hat{L}$ is obtained as shown in Equation (8).

$$L^\wedge = argmax\_c\ P(c \mid Q) \qquad (8)$$

The cognitive distance δ between the predicted level $L^\wedge$ and the target level L^* is then computed as the absolute difference between their ordinal positions in the Bloom's Taxonomy hierarchy, as shown in Equation (9).

$$\delta = |rank(L^\wedge) - rank(L*)| \qquad (9)$$

Based on the value of δ, the generated question is assigned one of three status categories, as defined in Table 7.

TABLE 7. Status classification based on cognitive distance

Status	Condition	Interpretation
Ideal	$\delta = 0$	Predicted level exactly matches target level
Slightly Ideal	$\delta = 1$	Predicted level is adjacent to target level
Non-Ideal	$\delta = 2$	Predicted level deviates significantly from target

Questions with status Ideal or Slightly Ideal are accepted as the final output and recorded along with their predicted level, confidence score, and number of attempts used. Questions with status Non-Ideal are rejected and the generation process in Phase 2 is repeated for the same topic τ and target level L^*. This loop runs for a maximum of three attempts. If an Ideal or Slightly Ideal question is produced before the limit is reached, the loop terminates early and that question is returned as the final output. If all three attempts are exhausted without producing an acceptable result, the question with the smallest δ across all attempts is selected as the final output. The complete result record for each question includes the question text, target level L^*, predicted level $L^\wedge$, cognitive distance δ, status category, confidence score, and total attempts used.

G. EVALUATION METRICS

This study employs two complementary evaluation schemes to assess system performance: classifier evaluation and paper-style evaluation. The classifier evaluation measures the intrinsic performance of the ModernBERT-Bloom-Taxonomy classifier on the Kaggle Bloom's Taxonomy dataset described in Section 3.2. Four metrics are computed per class using scikit-learn: Accuracy, Precision, Recall, and F1-score. Accuracy measures the proportion of correctly predicted instances out of all instances, as shown in Equation (10).

$$Accuracy = \frac{(TP + TN)}{N} \tag{10}$$

Precision measures the proportion of correctly predicted positive instances out of all instances predicted as positive, as shown in Equation (11).

$$Precision = \frac{TP}{(TP + FP)} \tag{11}$$

Recall measures the proportion of correctly predicted positive instances out of all actual positive instances, as shown in Equation (12).

$$Recall = \frac{TP}{(TP + FN)} \tag{12}$$

F1-score is the harmonic mean of Precision and Recall, providing a balanced measure when class distribution is uneven, as shown in Equation (13).

$$F1Score = 2 \times \frac{(Precision\ \times Recall)}{(Precision + Recall)} \tag{13}$$

Per-class metrics are aggregated as weighted averages across all six Bloom's Taxonomy levels, weighted by the number of true instances per class. A confusion matrix is additionally reported to visualize inter-class classification patterns across BT1–BT6.

The paper-style evaluation measures the end-to-end cognitive-level alignment of the full generation pipeline using the cognitive distance metric δ defined in previous section. In this scheme, TP, FP, and FN are redefined based on the value of δ rather than conventional classification correctness. A generated question is counted as a True Positive (TP) when $\delta = 0$, meaning the predicted level exactly matches the target level. A generated question is counted as a False Positive (FP) when $\delta = 1$, meaning the predicted level is one level adjacent to the target. A generated question is counted as a False Negative (FN) when $\delta \geq 2$, meaning the predicted level deviates significantly from the target. Using these definitions, Precision, Recall, F1, and Accuracy are computed using the same formulas as Equations (11), (12), (13), and a modified Accuracy as shown in Equation (14).

$$Accuracy = \frac{TP}{(TP + FP + FN)} \tag{14}$$

IV. RESULTS AND DISCUSSION

A. EXPERIMENTAL SETUP

All experiments were conducted on a local machine equipped with an NVIDIA GeForce RTX 3060 GPU (12 GB VRAM). The proposed framework was implemented in Python using the following libraries: LangChain for pipeline orchestration, Qdrant (in-memory) as the vector database, HuggingFace Transformers for the ModernBERT-Bloom-Taxonomy classifier, and Ollama for local LLM inference. The complete list of experimental parameters is summarized in Table 8.

TABLE 8. Experimental setup and parameters

Component	Configuration
GPU	NVIDIA GeForce RTX 3060 (12 GB VRAM)
Embedding Model	all-MiniLM-L6-v2 (384-dim, cosine)
Vector Database	Qdrant (in-memory)
Chunk Size	500 characters
Chunk Overlap	50 characters
Top-k Retrieval	3
Classifier	ModernBERT-Bloom-Taxonomy (HuggingFace)
LLM	Llama-3.1-8B, Qwen-3-8B, Gemma-3-4B, Mistral-7B (Ollama, local)
LLM Temperature	0.7
Max Regeneration Attempts	3
Total Topics Evaluated	319
Total Questions Generated	1,914

B. DATASET STATISTICS

This section presents descriptive statistics for both datasets used in the study. The OpenStax knowledge base dataset consists of 350 entries extracted and preprocessed from five subjects, subsequently split into 3,666 chunks for embedding and retrieval. Figure 3 presents the distribution of subject.

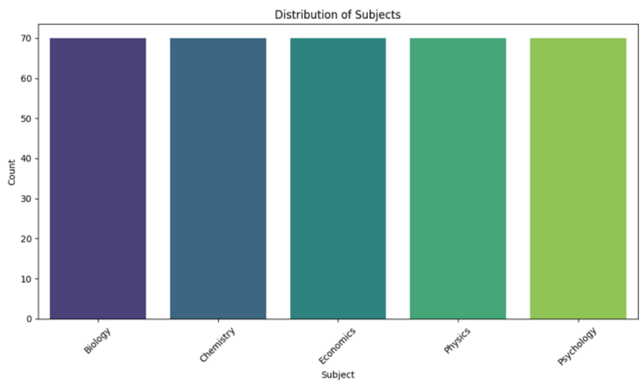

FIGURE 3. Distribution of Subjects OpenStax Learning Material

The Bloom's Taxonomy classification dataset sourced from Kaggle consists of 8,767 labeled questions distributed across six cognitive levels. Figure 4 presents the class distribution of this dataset.

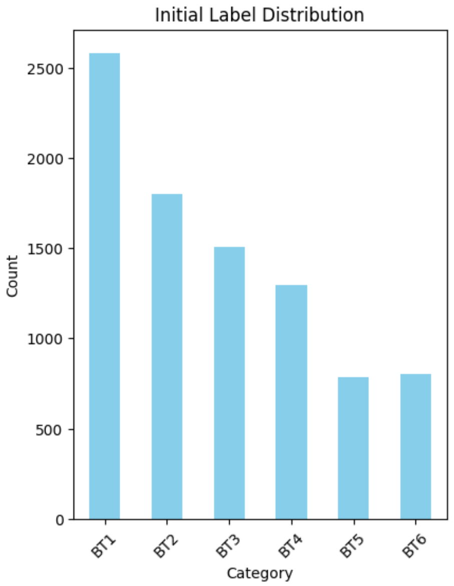

FIGURE 4. Label Distribution of the Bloom's Taxonomy Classification Dataset

As shown in Figure 4, the dataset exhibits a moderate class imbalance, with Remember (C1) being the most frequent class (29.45%) and Evaluate (C5) being the least frequent (8.93%). This imbalance is consistent with the natural distribution of educational questions, where lower-order cognitive tasks tend to be more prevalent. The weighted average metrics reported in the next section account for this imbalance.

C. CLASSIFIER EVALUATION RESULTS

The ModernBERT-Bloom-Taxonomy classifier was evaluated on the full Kaggle Bloom's Taxonomy dataset (n = 8,767) to assess its intrinsic performance as the cognitive-level verification component. The overall accuracy achieved was 86.36%, with a weighted precision of 0.8677, weighted recall of 0.8636, and weighted F1-score of 0.8629. The complete per-class results are presented in Table 9.

TABLE 9. Per-Class classifier evaluation results

Bloom's Level	C-Code	Precision	Recall	F1-Score
Remember	C1	0.79	0.91	0.85
Understand	C2	0.84	0.78	0.81
Apply	C3	0.88	0.89	0.88
Analyze	C4	0.91	0.74	0.82
Evaluate	C5	0.96	0.96	0.96
Create	C6	0.98	0.97	0.97
Macro Avg		0.89	0.87	0.88
Weighted Avg		0.87	0.86	0.86
Overall Acc				0.8636

The results indicate that the classifier performs strongly across most cognitive levels, with particularly high performance on higher-order levels. Create (C6) achieves the highest F1-score of 0.97, followed by Evaluate (C5) at 0.96. This is consistent with the linguistic characteristics of these levels, as questions targeting C5 and C6 typically contain distinct action verbs such as evaluate, justify, design, and construct that are relatively unambiguous for the classifier.

In contrast, Remember (C1) and Understand (C2) exhibit comparatively lower performance, with F1-scores of 0.85 and 0.81 respectively. The lower precision of Remember (0.79) suggests that the classifier tends to over-predict this class, likely because lower-order questions often use simple and general language that overlaps semantically with Understand-level questions. The lower recall of Analyze (C4) at 0.74 similarly indicates that the classifier frequently misclassifies Analyze-level questions as Understand or Apply, reflecting the linguistic proximity of these adjacent cognitive levels.

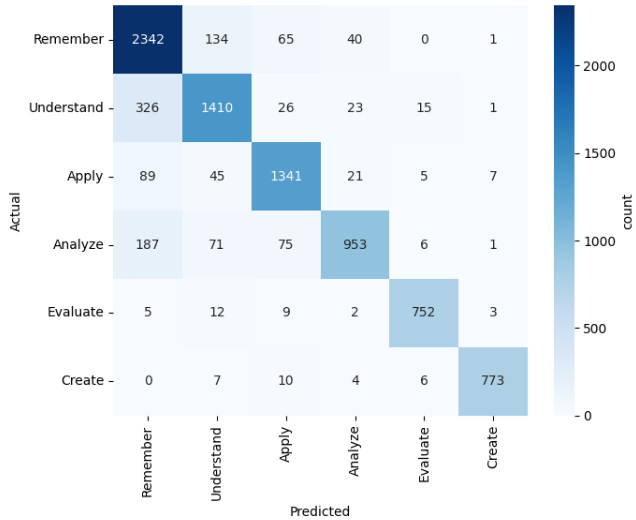

FIGURE 5. ModernBERT-Bloom-Taxonomy Confusion Matrix

These patterns are further confirmed by the confusion matrix (Fig. 5), which shows that the primary source of misclassification is between adjacent levels, particularly between C1 and C2 as well as between C3 and C4. This observation is consistent with the cognitive continuum nature of Bloom's Taxonomy, where neighboring levels share overlapping linguistic features [23]. It should be noted that ModernBERT-Bloom-Taxonomy is a pre-trained model adopted from a publicly available checkpoint; therefore, these results reflect the classifier's intrinsic capability rather than a direct contribution of this study. The reported accuracy of 86.36% confirms that the selected classifier is sufficiently reliable to serve as the cognitive-level verification component within the proposed generation pipeline.

D. QUESTION GENERATION RESULTS

This section presents the paper-style evaluation results for the end-to-end question generation pipeline across four LLMs:

TABLE 10. Overall paper-style evaluation results per LLM

LLM	TP (Ideal)	FP (Slightly Ideal)	FN (Non-Ideal)	Total	Precision	Recall	F1-score	Accuracy
Llama-3.1-8B	1,509	342	63	1,914	0.8152	0.9599	0.8817	0.7884
Mistral-7B	1,367	377	170	1,914	0.7838	0.8894	0.8333	0.7142
Qwen-3-8B	1,343	218	353	1,914	0.8603	0.7919	0.8247	0.7017
Gemma-3-4B	1,517	263	134	1,914	0.8522	0.9188	0.8843	0.7926

Llama-3.1-8B, Mistral-7B, Qwen-3-8B, and Gemma-3-4B. All models were evaluated under identical experimental conditions using the same RAG pipeline, prompt template, classifier, and feedback loop configuration described in proposed method. A total of 1,914 questions were generated per model (319 topics × 6 Bloom's levels). Table 10 presents the overall paper-style metrics for each LLM across all Bloom's levels.

Gemma-3-4B achieves the highest number of Ideal questions (1,517) and the highest overall Accuracy (0.7926), while Llama-3.1-8B achieves the highest F1-score (0.8817). Qwen-3-8B attains the highest Precision (0.8603), indicating fewer Slightly Ideal predictions relative to Ideal ones, but at the cost of a substantially higher Non-Ideal count (353), resulting in the lowest Accuracy (0.7017). Mistral-7B performs consistently across most metrics but does not lead in any single category. Tables 11 through 14 present the paper-style metrics broken down per Bloom's Level (BL) for each LLM, including the underlying TP (Ideal), FP (Slightly Ideal), and FN (Non-Ideal) counts that determine each derived metric. Reporting the raw counts in addition to the derived metrics allows the reader to inspect the specific direction in which each model deviates from the target cognitive level: a high FP count indicates predominantly adjacent-level drift, whereas a high FN count indicates substantial cognitive-level deviation. With n = 319 per level (319 topics × one final-attempt question), the three counts always sum to the row total, providing full reproducibility for every derived metric.

TABLE 11. Llama-3.1-8B per-level paper-style results

BL	TP	FP	FN	Precision	Recall	F1	Accuracy
C1	295	13	11	0.95	0.96	0.96	0.92
C2	225	77	17	0.74	0.92	0.82	0.70
C3	258	56	5	0.82	0.98	0.89	0.80
C4	291	19	9	0.93	0.97	0.95	0.91
C5	130	177	12	0.42	0.91	0.57	0.40
C6	310	0	9	1.00	0.97	0.98	0.97

TABLE 12. Mistral-7B per-level paper-style results

BL	TP	FP	FN	Precision	Recall	F1	Accuracy
C1	298	13	8	0.95	0.97	0.96	0.93
C2	240	67	12	0.78	0.95	0.85	0.75
C3	163	154	2	0.51	0.98	0.67	0.51
C4	201	29	89	0.87	0.69	0.77	0.63
C5	195	106	18	0.64	0.91	0.75	0.61
C6	270	8	41	0.97	0.86	0.91	0.84

TABLE 13. Qwen-3-8B per-level paper-style results

BL	TP	FP	FN	Precision	Recall	F1	Accuracy
C1	289	13	17	0.95	0.94	0.95	0.90
C2	312	6	1	0.98	0.99	0.98	0.97
C3	190	113	16	0.62	0.92	0.74	0.59
C4	84	23	212	0.78	0.28	0.41	0.26
C5	228	62	29	0.78	0.88	0.83	0.71
C6	240	1	78	0.99	0.75	0.85	0.75

TABLE 14. Gemma-3-4B per-level paper-style results

BL	TP	FP	FN	Precision	Recall	F1	Accuracy
C1	267	10	42	0.96	0.86	0.91	0.83
C2	302	13	4	0.95	0.98	0.97	0.94
C3	163	131	25	0.55	0.86	0.67	0.51
C4	253	30	36	0.89	0.87	0.88	0.79
C5	227	77	15	0.74	0.93	0.83	0.71
C6	305	2	12	0.99	0.96	0.97	0.95

Table 15 summarizes the F1-score per Bloom's level across all four LLMs for direct comparison. F1-score is used in this context as the primary per-level metric because it provides a balanced measure between Precision and Recall, capturing both the ability of the pipeline to generate Ideal questions and its ability to avoid Non-Ideal outputs simultaneously. A high

F1-score at a given level indicates that the LLM consistently produces questions whose predicted cognitive level exactly matches the target level, while a low F1-score reflects either a high rate of Non-Ideal generation or a tendency to produce Slightly Ideal questions at adjacent levels.

The table reveals that no single LLM dominates across all six levels simultaneously, indicating that cognitive-level alignment is highly dependent on both the model's instruction-following capability and the linguistic characteristics of each Bloom's level. Llama-3.1-8B achieves the highest F1-score at four levels (C1, C3, C4, C6), making it the most consistently strong model overall. Qwen-3-8B leads at C2 (Understand) with an F1-score of 0.98 but collapses at C4 (Analyze) with only 0.41, exhibiting the highest inter-level variance among all models. Gemma-3-4B demonstrates the most balanced profile for higher-order levels (C2 through C5), while Mistral-7B shows moderate performance without leading at any single level. These patterns suggest that the difficulty of cognitive-level generation does not follow a monotonically increasing trend with Bloom's level, but rather reflects the degree of linguistic overlap between adjacent levels, particularly between C2 and C3, and between C4 and C5, which varies in sensitivity across different LLM architectures.

TABLE 15. F1-score per bloom's level per LLM

Bloom's Level	Llama-3.1-8B	Mistral-7B	Qwen-3-8B	Gemma-3-4B
Remember	0.96	0.96	0.95	0.91
Understand	0.82	0.85	0.98	0.97
Apply	0.89	0.67	0.74	0.67
Analyze	0.95	0.77	0.41	0.88
Evaluate	0.57	0.75	0.83	0.83
Create	0.98	0.91	0.85	0.97

E. DISCUSSION

Gemma-3-4B and Llama-3.1-8B emerge as the two strongest models overall, with Accuracy of 0.7926 and 0.7884 and F1-scores of 0.88 and 0.88 respectively. The performance gap between these two models and the remaining two is relatively small in terms of F1-score, but more pronounced in Accuracy, reflecting the ability of Gemma-3-4B and Llama-3.1-8B to produce a higher proportion of exact cognitive-level matches (Ideal) relative to near-matches (Slightly Ideal) and failures (Non-Ideal).

Qwen-3-8B achieves the highest Precision (0.86) among all models, indicating that when it does produce an Ideal question, it is rarely accompanied by Slightly Ideal outputs. However, its substantially high Non-Ideal count (353), more than five times that of Llama-3.1-8B (63), suggests that Qwen-3-8B tends to either produce exactly correct or significantly off-target questions, with less tolerance for adjacent-level generation. This behavior is most critically observed at C4 (Analyze), where Qwen-3-8B achieves an F1-score of only 0.41 with 212 Non-Ideal questions, representing the worst single-level result across all models, and suggesting that Qwen-3-8B struggles to internalize the linguistic distinction between Analyze-level and lower-order cognitive tasks within the prompt-guided generation setting. In contrast, Mistral-7B shows moderate and relatively balanced performance across most levels, with its primary weakness at C3 (Apply), where a high FP count of 154 indicates that many Apply-level questions are generated at an adjacent level, suggesting that Mistral-7B frequently interprets Apply-level instructions in the prompt as Understand-level tasks.

Examining individual Bloom's levels reveals consistent patterns across the four models. C1 (Remember) is consistently the highest-performing level across all models, with F1-scores ranging from 0.9113 (Gemma-3-4B) to 0.96 (Mistral-7B). This is expected, as Remember-level questions typically use simple and unambiguous recall-based language that is straightforwardly distinguishable by both the LLM and the classifier. This result is consistent with prior studies that characterize Remember as the lowest cognitive level in Bloom's Taxonomy, requiring primarily factual recall and recognition rather than higher-order reasoning processes [24].

At C2 (Understand), the results show high variability across models. Qwen-3-8B and Gemma-3-4B achieve F1-scores of 0.98 and 0.97 respectively, whereas Llama-3.1-8B achieves only 0.82. The lower Llama performance at this level is attributable to a high FP count (77), indicating that Llama-3.1-8B frequently generates questions that the classifier predicts as C1 (Remember) rather than C2 (Understand), a consistent confusion pattern also observed in the classifier evaluation.

C3 (Apply) is the most problematic level for three of the four models. Mistral-7B and Gemma-3-4B both achieve F1-scores of 0.67, while Qwen-3-8B achieves 0.74. Only Llama-3.1-8B achieves a substantially higher F1-score of 0.89 at this level. The high FP counts across models suggest that Apply-level prompts tend to elicit questions that are classified as Understand-level, reflecting the linguistic overlap between these two adjacent cognitive levels. This confusion is consistent with prior Bloom's Taxonomy studies reporting substantial semantic overlap between Understand and Apply categories, where questions often differ only in the degree of contextual application rather than linguistic form [23].

C4 (Analyze) exhibits the most dramatic inter-model disparity. Llama-3.1-8B achieves 0.95 and Gemma-3-4B achieves 0.88, while Qwen-3-8B collapses to 0.4169 with 212 Non-Ideal results. This extreme failure of Qwen-3-8B at C4 suggests a systematic tendency to generate questions at substantially lower cognitive levels when targeting Analyze, possibly due to differences in how the model interprets analytical action verbs such as analyze, contrast, and differentiate within the prompt context. Prior studies have reported that Analyze-level questions are among the most difficult categories to identify automatically because analytical reasoning is often expressed through linguistic patterns that overlap with lower cognitive levels, resulting in frequent misclassification and generation drift [23] [25].

TABLE 16. Comparison of proposed framework results against prior work

Aspect	Aisyah et al. [5]	Banujan et al. [6]	Scaria et al. [12]	Proposed Work
Question Generation	✓	✗	✓	✓
Bloom's Taxonomy (C1–C6)	✗	✓	✓ (prompt-guided)	✓ (classifier-verified)
Cognitive Verification	✗	✓ (existing questions)	✗	✓ (generated questions)
Feedback Loop	✗	✗	✗	✓
RAG-based Retrieval	✓	✗	✗	✓
Multi-LLM Comparison	✓	✗	✓	✓
Classifier Accuracy	N/A	84.2% (LSTM+BERT)	N/A	86.36% (ModernBERT)
Cognitive Alignment Metric	None	N/A	Qualitative	Ideal Rate 78.84–79.26%
Best Performing Model	Llama-3-8B	LSTM+BERT	GPT-4	Llama-3.1-8B / Gemma-3-4B

C5 (Evaluate) is the weakest level for Llama-3.1-8B (F1 = 0.57), driven primarily by a very high FP count of 177, meaning that the majority of generated questions classified as non-Ideal at this level are Slightly Ideal rather than completely off-target. This indicates that Llama-3.1-8B tends to generate Evaluate-level questions that the classifier places at C4 (Analyze) or C6 (Create), suggesting proximity between the LLM's interpretation of Evaluate and adjacent higher-order levels. By contrast, Mistral-7B (0.75), Qwen-3-8B (0.83), and Gemma-3-4B (0.83) all perform notably better at C5, with Qwen-3-8B achieving the highest F1-score at this level.

C6 (Create) is consistently strong across all models, with Llama-3.1-8B achieving the highest F1-score of 0.98 followed by Gemma-3-4B at 0.97. Notably, Llama-3.1-8B achieves a perfect Precision of 1.00 at C6, meaning that all of its generated Create-level questions that were not Non-Ideal were classified as exactly Ideal, with zero Slightly Ideal outputs at this level.

Taken together, Llama-3.1-8B demonstrates the most balanced performance profile, achieving the highest F1-score at four out of six Bloom's levels (C1, C3, C4, C6) and the best overall F1-score of 0.88. Its primary weakness lies at C5 (Evaluate), where a large proportion of generated questions are classified as adjacent levels. Gemma-3-4B follows closely with the highest overall Accuracy and strong performance across all levels, particularly at C2 and C6. Qwen-3-8B excels at C2 but exhibits a critical failure at C4, making it unsuitable as a standalone generation model without additional prompt tuning. Mistral-7B provides moderate and consistent performance but does not lead in any single metric, suggesting it is the least effective model within this pipeline for cognitive-level question generation. These results confirm that the choice of LLM has a significant impact on cognitive-level alignment quality, and that Llama-3.1-8B represents the most suitable model for the proposed RAG-based Bloom's Taxonomy question generation framework under the current experimental conditions.

The results of the proposed framework can be contextualized against prior work reviewed in related work section. Table 16 summarizes the key distinctions between this study and the most closely related prior studies across several capability dimensions.

V. CONCLUSION

This study addressed the fundamental limitation of existing automatic question generation approaches that produce questions without guaranteed alignment to specific cognitive levels of Bloom's Taxonomy. By integrating Retrieval Augmented Generation, Large Language Models, and a pretrained ModernBERT Bloom Taxonomy classifier within a unified pipeline, the proposed framework directly responds to the gap identified in prior work, in which question generation and cognitive level classification have operated as decoupled processes. The system combines contextual retrieval from a curated OpenStax knowledge base, prompt guided generation conditioned on a target Bloom's level, and classifier based verification with a regenerative feedback loop, allowing the production of contextually relevant questions whose cognitive complexity can be explicitly measured against the six levels of Revised Bloom's Taxonomy from Remember (C1) through Create (C6), thereby replacing the coarse low, medium, and high difficulty categories used in earlier RAG based studies.

The experimental results across 1,914 generated questions per model and four representative open weight Large Language Models confirm the viability of the proposed framework. The ModernBERT classifier achieved an overall accuracy of 86.36 percent on the Kaggle Bloom's Taxonomy dataset, establishing a reliable cognitive verification component within the pipeline. At the end to end level, Gemma-3-4B obtained the highest proportion of cognitively ideal questions with an accuracy of 79 percent, while Llama 3.1 8B achieved the strongest overall F1 score of 0.88 and led at four of the six Bloom's levels, namely Remember, Apply, Analyze, and Create. Qwen-3-8B attained the highest precision but exhibited a critical failure at the Analyze level, whereas Mistral-7B demonstrated moderate yet balanced performance. The regenerative feedback loop ensured that questions initially deviating from the target level were iteratively refined toward acceptable cognitive alignment within a maximum of three attempts. These findings collectively demonstrate that the framework fulfills its stated contributions, including the integration of generation, retrieval, and verification components, the use of explicit C1 to C6 classification rather than general difficulty labels, the implementation of a regenerative feedback mechanism, the generation of contextually grounded questions with measurable cognitive alignment, and the provision of a reusable building block for adaptive assessment in Learning Management Systems.

The analysis further revealed that cognitive level alignment is sensitive to both the linguistic overlap between adjacent Bloom's levels and the instruction following behavior of individual Large Language Models, with the most consistent confusion patterns occurring between C1 and C2 as well as between C3 and C4. This sensitivity highlights that no single Large Language Model dominates across all cognitive levels, and that the choice of generator should be informed by the target distribution of cognitive complexity required in a given educational context. The framework therefore positions itself not as a replacement for human expert review, but as a scalable assistive mechanism that produces verifiable assessment items at scale while preserving pedagogical interpretability.

Several directions remain open for future research. The framework can be extended to multilingual settings, particularly for educational systems operating outside English speaking contexts, by incorporating multilingual embedding models and multilingual variants of the cognitive classifier. Further fine tuning of the ModernBERT classifier on domain specific question banks may reduce the residual misclassification observed between adjacent cognitive levels and improve overall alignment quality. The regenerative feedback loop can also be enhanced through reinforcement learning or preference optimization techniques, enabling the Large Language Model to learn directly from the classifier's verification signal rather than relying solely on prompt resampling. Expanding the knowledge base beyond the five OpenStax subjects to encompass broader academic and professional domains would strengthen the generalizability and domain adaptability of the system, and future work will additionally involve human expert validation studies to compare the cognitive alignment judgments of educators with the automated classifier predictions.

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The preferred spelling of the word “acknowledgment” in American English is without an “e” after the “g.” Use the singular heading even if you have many acknowledgments. Avoid expressions such as “One of us (S.B.A.).

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